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Project Phase 2 (Term 252)

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ICS-445: Network Management and Security

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Overview and Objectives

This section outlines the configurations applied to the network topology to establish proper routing and high availability. The configurations performed in this section successfully accomplished the following assignment requirements:

- Configured IPv4 addressing on routers and switches.
- Enabled Layer 3 routing on multilayer switches.
- Configured Layer-3 EtherChannel between core switches.
- Configured management VLAN IP addresses.
- Configured HSRPv2 for all required VLANs.
- Set proper priorities and enabled preemption as specified.

By completing these tasks, the network now ensures that inter-VLAN communication functions correctly, HSRP active and standby roles are correctly assigned , and end-user default gateways operate seamlessly through the virtual IP addresses.

Device Configurations and Screenshots

Below are the clear screenshots showing all the steps and commands implemented on each device to achieve the objectives of Part 3.

1. Router (R1)

The screenshot below demonstrates the IP addressing and routing configurations applied to R1.

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface range g0/0/0,g0/1/0
R1(config-if-range)#ip address dhcp
R1(config-if-range)#no shutdown

R1(config-if-range)#interface g0/0
R1(config-if)#ip address 10.0.0.33 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#interface g0/1
R1(config-if)#ip address 10.0.0.37 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#interface loopback0
R1(config-if)#ip address 10.0.0.76 255.255.255.255
R1(config-if)#exit
R1(config)#do write
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

R1(config)#do write
%DHCP-6-ADDRESS_ASSIGN: Interface GigabitEthernet0/1/0 assigned DHCP address 203.0.113.6,
mask 255.255.255.252, hostname R1
```

2. Core Switches (CSW1 & CSW2)

The screenshots below display the Layer 3 routing enablement, Layer-3 EtherChannel configuration between the core switches, and relevant IP assignments.

- **CSW1:**

```
CSW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
CSW1(config)#ip routing
CSW1(config)#interface range g1/0/2-3
CSW1(config-if-range)#no switchport
CSW1(config-if-range)#channel-group 1 mode desirable
CSW1(config-if-range)#exit
CSW1(config)#do write
%LINK-3-UPDOWN: Interface Port-channel1, changed state to down
```

```
CSW1(config)#interface po1
CSW1(config-if)#ip address 10.0.0.41 255.255.255.252
CSW1(config-if)#interface g1/0/1
CSW1(config-if)#no switchport
CSW1(config-if)#ip address 10.0.0.34 255.255.255.252
CSW1(config-if)#interface g1/1/1
CSW1(config-if)#no switchport
CSW1(config-if)#ip address 10.0.0.45 255.255.255.252
CSW1(config-if)#interface g1/1/2
CSW1(config-if)#no switchport
CSW1(config-if)#ip address 10.0.0.49 255.255.255.252
CSW1(config-if)#interface g1/1/3
CSW1(config-if)#no switchport
CSW1(config-if)#ip address 10.0.0.53 255.255.255.252
CSW1(config-if)#interface g1/1/4
CSW1(config-if)#no switchport
CSW1(config-if)#ip address 10.0.0.57 255.255.255.252
CSW1(config-if)#interface loopback0
CSW1(config-if)#ip address 10.0.0.77 255.255.255.255
CSW1(config-if)#interface range g1/0/4-24
CSW1(config-if-range)#shutdown
CSW1(config-if-range)#exit
```

- **CSW2:**

```
Enter configuration commands, one per line.  End with CNTL/Z.
CSW2(config)#ip routing
CSW2(config)#interface range g1/0/2-3
CSW2(config-if-range)#no switchport
CSW2(config-if-range)#channel-group 1 mode desirable
CSW2(config-if-range)#exit
CSW2(config)#do write
%LINK-3-UPDOWN: Interface Port-channel1, changed state to down
```

```
CSW2(config)#interface po1
CSW2(config-if)#ip address 10.0.0.42 255.255.255.252
CSW2(config-if)#interface g1/0/1
CSW2(config-if)#no switchport
CSW2(config-if)#ip address 10.0.0.38 255.255.255.252
CSW2(config-if)#interface g1/1/1
CSW2(config-if)#no switchport
CSW2(config-if)#ip address 10.0.0.61 255.255.255.252
CSW2(config-if)#interface g1/1/2
CSW2(config-if)#no switchport
CSW2(config-if)#ip address 10.0.0.65 255.255.255.252
CSW2(config-if)#interface g1/1/3
CSW2(config-if)#no switchport
CSW2(config-if)#ip address 10.0.0.69 255.255.255.252
CSW2(config-if)#interface g1/1/4
CSW2(config-if)#no switchport
CSW2(config-if)#ip address 10.0.0.73 255.255.255.252
CSW2(config-if)#interface loopback0
CSW2(config-if)#ip address 10.0.0.78 255.255.255.255
CSW2(config-if)#interface range g1/0/4-24
CSW2(config-if-range)#shutdown
CSW2(config-if-range)#exit
CSW2(config)#do write
```

3. Distribution Switches - Block A (DSW-A1 & DSW-A2)

The following screenshots capture the configuration of management VLAN IP addresses, HSRPv2 setup for required VLANs, and the assignment of priorities and preemption.

- **DSW-A1:**

```
DSW-A1(config)#ip routing
DSW-A1(config)#interface g1/1/1
DSW-A1(config-if)#no switchport
DSW-A1(config-if)#ip address 10.0.0.46 255.255.255.252
DSW-A1(config-if)#interface g1/1/2
DSW-A1(config-if)#no switchport
DSW-A1(config-if)#ip address 10.0.0.62 255.255.255.252
DSW-A1(config-if)#interface loopback0
DSW-A1(config-if)#ip address 10.0.0.79 255.255.255.255
DSW-A1(config-if)#interface vlan 99
DSW-A1(config-if)#ip address 10.0.0.2 255.255.255.240
DSW-A1(config-if)#standby version 2
DSW-A1(config-if)#standby 1 ip 10.0.0.1
DSW-A1(config-if)#standby 1 priority 105
DSW-A1(config-if)#standby 1 preempt
DSW-A1(config-if)#interface vlan 10
DSW-A1(config-if)#ip address 10.1.0.2 255.255.255.0
DSW-A1(config-if)#standby version 2
DSW-A1(config-if)#standby 2 ip 10.1.0.1
DSW-A1(config-if)#standby 2 priority 105
DSW-A1(config-if)#standby 2 preempt
DSW-A1(config-if)#interface vlan 20
DSW-A1(config-if)#ip address 10.2.0.2 255.255.255.0
DSW-A1(config-if)#standby version 2
DSW-A1(config-if)#standby 3 ip 10.2.0.1
DSW-A1(config-if)#interface vlan 40
DSW-A1(config-if)#ip address 10.6.0.2 255.255.255.0
DSW-A1(config-if)#standby version 2
```

```
DSW-A1(config-if)#standby 4 ip 10.6.0.1
DSW-A1(config-if)#exit
DSW-A1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
DSW-A1(config)#
%HSRP-6-STATECHANGE: Vlan10 Grp 2 state Standby -> Active

DSW-A1(config)#
%HSRP-6-STATECHANGE: Vlan20 Grp 3 state Speak -> Standby

DSW-A1(config)#
%HSRP-6-STATECHANGE: Vlan99 Grp 1 state Speak -> Standby
```

- **DSW-A2:**

```
DSW-A2(config)#ip routing
DSW-A2(config)#interface g1/1/1
DSW-A2(config-if)#no switchport
DSW-A2(config-if)#ip address 10.0.0.50 255.255.255.252
DSW-A2(config-if)#interface g1/1/2
DSW-A2(config-if)#no switchport
DSW-A2(config-if)#ip address 10.0.0.66 255.255.255.252
DSW-A2(config-if)#interface loopback0
DSW-A2(config-if)#ip address 10.0.0.80 255.255.255.255
DSW-A2(config-if)#interface vlan 99
DSW-A2(config-if)#ip address 10.0.0.3 255.255.255.240
DSW-A2(config-if)#standby version 2
DSW-A2(config-if)#standby 1 ip 10.0.0.1
DSW-A2(config-if)#interface vlan 10
DSW-A2(config-if)#ip address 10.1.0.3 255.255.255.0
DSW-A2(config-if)#standby version 2
DSW-A2(config-if)#standby 2 ip 10.1.0.1
DSW-A2(config-if)#interface vlan 20
DSW-A2(config-if)#ip address 10.2.0.3 255.255.255.0
DSW-A2(config-if)#standby version 2
DSW-A2(config-if)#standby 3 ip 10.2.0.1
DSW-A2(config-if)#standby 3 priority 105
DSW-A2(config-if)#standby 3 preempt
DSW-A2(config-if)#interface vlan 40
DSW-A2(config-if)#ip address 10.6.0.3 255.255.255.0
DSW-A2(config-if)#standby version 2
DSW-A2(config-if)#standby 4 ip 10.6.0.1
DSW-A2(config-if)#standby 4 priority 105
DSW-A2(config-if)#standby 4 preempt
DSW-A2(config-if)#exit
DSW-A2(config)#do write
```

4. Distribution Switches - Block B (DSW-B1 & DSW-B2)

The following screenshots capture the configuration of management VLAN IP addresses, HSRPv2 setup for required VLANs, and the assignment of priorities and preemption.

- **DSW-B1:**

```
DSW-B1(config)#ip routing
DSW-B1(config)#interface g1/1/1
DSW-B1(config-if)#no switchport
DSW-B1(config-if)#ip address 10.0.0.54 255.255.255.252
DSW-B1(config-if)#interface g1/1/2
DSW-B1(config-if)#no switchport
DSW-B1(config-if)#ip address 10.0.0.70 255.255.255.252
DSW-B1(config-if)#interface loopback0
DSW-B1(config-if)#ip address 10.0.0.81 255.255.255.255
DSW-B1(config-if)#interface vlan 99
DSW-B1(config-if)#ip address 10.0.0.18 255.255.255.240
DSW-B1(config-if)#standby version 2
DSW-B1(config-if)#standby 1 ip 10.0.0.17
DSW-B1(config-if)#standby 1 priority 105
DSW-B1(config-if)#standby 1 preempt
DSW-B1(config-if)#interface vlan 10
DSW-B1(config-if)#ip address 10.3.0.2 255.255.255.0
DSW-B1(config-if)#standby version 2
DSW-B1(config-if)#standby 2 ip 10.3.0.1
DSW-B1(config-if)#standby 2 priority 105
DSW-B1(config-if)#standby 2 preempt
DSW-B1(config-if)#interface vlan 20
DSW-B1(config-if)#ip address 10.4.0.2 255.255.255.0
DSW-B1(config-if)#standby version 2
DSW-B1(config-if)#standby 3 ip 10.4.0.1
DSW-B1(config-if)#interface vlan 30
DSW-B1(config-if)#ip address 10.5.0.2 255.255.255.0
DSW-B1(config-if)#standby version 2
DSW-B1(config-if)#standby 4 ip 10.5.0.1
DSW-B1(config-if)#exit
DSW-B1(config)#do write
```

- **DSW-B2:**

```
DSW-B2(config)#ip routing
DSW-B2(config)#interface g1/1/1
DSW-B2(config-if)#no switchport
DSW-B2(config-if)#ip address 10.0.0.58 255.255.255.252
DSW-B2(config-if)#interface g1/1/2
DSW-B2(config-if)#no switchport
DSW-B2(config-if)#ip address 10.0.0.74 255.255.255.252
DSW-B2(config-if)#interface loopback0
DSW-B2(config-if)#ip address 10.0.0.82 255.255.255.255
DSW-B2(config-if)#interface vlan 99
DSW-B2(config-if)#ip address 10.0.0.19 255.255.255.240
DSW-B2(config-if)#standby version 2
DSW-B2(config-if)#standby 1 ip 10.0.0.17
DSW-B2(config-if)#interface vlan 10
DSW-B2(config-if)#ip address 10.3.0.3 255.255.255.0
DSW-B2(config-if)#standby version 2
DSW-B2(config-if)#standby 2 ip 10.3.0.1
DSW-B2(config-if)#interface vlan 20
DSW-B2(config-if)#ip address 10.4.0.3 255.255.255.0
DSW-B2(config-if)#standby version 2
DSW-B2(config-if)#standby 3 ip 10.4.0.1
DSW-B2(config-if)#standby 3 priority 105
DSW-B2(config-if)#standby 3 preempt
DSW-B2(config-if)#interface vlan 30
DSW-B2(config-if)#ip address 10.5.0.3 255.255.255.0
DSW-B2(config-if)#standby version 2
DSW-B2(config-if)#standby 4 ip 10.5.0.1
DSW-B2(config-if)#standby 4 priority 105
DSW-B2(config-if)#standby 4 preempt
DSW-B2(config-if)#exit
DSW-B2(config)#do write
```

5. Access Switches - Block A (ASW-A1, ASW-A2, & ASW-A3)

The screenshots below detail the IPv4 addressing and management VLAN configurations applied to the Block A access layer switches.

- **ASW-A1:**

```
ASW-A1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-A1(config)#ip default-gateway 10.0.0.1
ASW-A1(config)#interface vlan 99
ASW-A1(config-if)#ip address 10.0.0.4 255.255.255.240
ASW-A1(config-if)#exit
ASW-A1(config)#do write
Building configuration...
[OK]
```

- **ASW-A2:**

```
ASW-A2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-A2(config)#ip default-gateway 10.0.0.1
ASW-A2(config)#interface vlan 99
ASW-A2(config-if)#ip address 10.0.0.5 255.255.255.240
ASW-A2(config-if)#exit
ASW-A2(config)#do write
Building configuration...
[OK]
```

- **ASW-A3:**

```
ASW-A3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-A3(config)#ip default-gateway 10.0.0.1
ASW-A3(config)#interface vlan 99
ASW-A3(config-if)#ip address 10.0.0.6 255.255.255.240
ASW-A3(config-if)#exit
ASW-A3(config)#do write
Building configuration...
[OK]
```

6. Access Switches - Block B (ASW-B1, ASW-B2, & ASW-B3)

The screenshots below detail the IPv4 addressing and management VLAN configurations applied to the Block B access layer switches.

- **ASW-B1:**

```
ASW-B1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-B1(config)#ip default-gateway 10.0.0.17
ASW-B1(config)#interface vlan 99
ASW-B1(config-if)#ip address 10.0.0.20 255.255.255.240
ASW-B1(config-if)#exit
ASW-B1(config)#do write
Building configuration...
[OK]
```

- **ASW-B2:**

```
ASW-B2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-B2(config)#
ASW-B2(config)#ip default-gateway 10.0.0.17
ASW-B2(config)#interface vlan 99
ASW-B2(config-if)#ip address 10.0.0.21 255.255.255.240
ASW-B2(config-if)#exit
ASW-B2(config)#do write
Building configuration...
[OK]
```

- **ASW-B3:**

```
ASW-B3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-B3(config)#ip default-gateway 10.0.0.17
ASW-B3(config)#interface vlan 99
ASW-B3(config-if)#ip address 10.0.0.22 255.255.255.240
ASW-B3(config-if)#exit
ASW-B3(config)#do write
Building configuration...
[OK]
```

Part 4: Rapid Spanning Tree Protocol (RSTP) Configuration

This section outlines the configurations applied to the network topology to implement Rapid Spanning Tree Protocol (RSTP) and ensure optimal Layer-2 redundancy and loop prevention. The configurations performed in this section successfully accomplished the following assignment requirements:

- Configuring Rapid PVST+ on all Access and Distribution switches
- Aligning the STP root bridge with the HSRP active router for each VLAN
- Configuring STP priorities correctly
- Enabling PortFast on end-host ports
- Enabling BPDU Guard on end-host ports
- By completing these tasks, the network now ensures that Layer-2 loops are prevented, convergence occurs rapidly after topology changes, and traffic flows through the most efficient paths aligned with the HSRP active gateways. This design enhances network stability, redundancy, and overall performance across the switching infrastructure.

Below are the clear screenshots showing all the steps and commands implemented on each device to achieve the objectives of Part 4.

1. **Distribution Switches – Block A (DSW-A1 & DSW-A2)**
   The screenshots below demonstrate the Rapid PVST+ configuration and the assignment of spanning-tree priorities to control root bridge election for the VLANs in Block A.

- **DSW-A1:**
  The configuration shows that Rapid PVST+ is enabled and the switch is configured as the primary root bridge for VLANs 10 and 99 by assigning a priority of 0. VLANs 20 and 40 are assigned a higher priority of 4096, allowing another distribution switch to become the root bridge for those VLANs.

```
DSW-A1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-A1(config)#spanning-tree mode rapid-pvst
DSW-A1(config)#spanning-tree vlan 10,99 priority 0
DSW-A1(config)#spanning-tree vlan 20,40 priority 4096
DSW-A1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

- **DSW-A2:**
  The configuration shows Rapid PVST+ enabled on the switch. The switch is configured as the primary root bridge for VLANs 20 and 40 with priority 0, while VLANs 10 and 99 are assigned priority 4096, making this switch the secondary root bridge for those VLANs.

  This design aligns the STP root bridge with the HSRP active router and balances Layer-2 traffic between distribution switches.

```
DSW-A2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-A2(config)#spanning-tree mode rapid-pvst
DSW-A2(config)#
%HSRP-6-STATECHANGE: Vlan10 Grp 2 state Standby -> Active

DSW-A2(config)#spanning-tree vlan 10,99 priority 4096
DSW-A2(config)#spanning-tree vlan 20,40 priority 0
DSW-A2(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

2. **Distribution Switches – Block B (DSW-B1 & DSW-B2)**
   The screenshots below demonstrate the Rapid PVST+ configuration and the spanning-tree priority assignments for Block B distribution switches.

- **DSW-B1:**

  Rapid PVST+ is enabled and the switch is configured as the primary root bridge for VLANs 10 and 99 by assigning the lowest priority value (0). VLANs 20 and 30 are configured with priority 4096, allowing another switch to take the primary role for those VLANs.

```
DSW-B1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-B1(config)#spanning-tree mode rapid-pvst
DSW-B1(config)#spanning-tree vlan 10,99 priority 0
DSW-B1(config)#spanning-tree vlan 20,30 priority 4096
DSW-B1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

- **DSW-B2:**

  Rapid PVST+ is enabled, and the switch is configured as the primary root bridge for VLANs 20 and 30 with priority 0. VLANs 10 and 99 are configured with priority 4096, making them the secondary root bridges for those VLANs.

  This configuration provides redundancy and load balancing across the distribution layer.

```
DSW-B2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-B2(config)#spanning-tree mode rapid-pvst
DSW-B2(config)#spanning-tree vlan 10,99 priority 4096
DSW-B2(config)#spanning-tree vlan 20,30 priority 0
DSW-B2(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

3. **Access Switches – Block A (ASW-A1, ASW-A2 & ASW-A3)**
   The screenshots below demonstrate the Rapid PVST+ configuration and the implementation of PortFast and BPDU Guard on access ports connected to end-user devices.

- **ASW-A1:**
  Rapid PVST+ is enabled on the switch. PortFast is configured on the access interfaces to allow devices to transition immediately to the forwarding state without waiting for spanning-tree convergence. BPDU Guard is also enabled to automatically disable the port if BPDUs are detected, protecting the network from potential switching loops.

```
ASW-A1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-A1(config)#spanning-tree mode rapid-pvst
ASW-A1(config)#interface f0/1
ASW-A1(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
ASW-A1(config-if)#spanning-tree bpduguard enable
ASW-A1(config-if)#exit
ASW-A1(config)#do write
Building configuration...
[OK]
ASW-A1(config)#interface f0/2
ASW-A1(config-if)#spanning-tree portfast trunk
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
ASW-A1(config-if)#spanning-tree bpduguard enable
ASW-A1(config-if)#exit
ASW-A1(config)#do write
Building configuration...
[OK]
```

- **ASW-A2:**
  Rapid PVST+ is enabled and PortFast is configured on the host-facing interfaces. BPDU Guard is enabled to protect the network from accidental switch connections or misconfigured devices.

```
ASW-A2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-A2(config)#spanning-tree mode rapid-pvst
ASW-A2(config)#interface f0/1
ASW-A2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
ASW-A2(config-if)#spanning-tree bpduguard enable
ASW-A2(config-if)#exit
ASW-A2(config)#do write
Building configuration...
```

- **ASW-A3:**
  Rapid PVST+ is enabled with PortFast and BPDU Guard configured on the access interfaces. This ensures faster host connectivity and prevents unauthorized switching devices from creating Layer-2 loops.

```
ASW-A3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-A3(config)#spanning-tree mode rapid-pvst
ASW-A3(config)#interface f0/1
ASW-A3(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
ASW-A3(config-if)#spanning-tree bpduguard enable
ASW-A3(config-if)#exit
ASW-A3(config)#do write
Building configuration...
[OK]
```

4. **Access Switches – Block B (ASW-B1, ASW-B2 & ASW-B3)**
   The screenshots below demonstrate the Rapid PVST+ configuration and the protection mechanisms implemented on Block B access switches.

- **ASW-B1**
  Rapid PVST+ is enabled and PortFast is configured on access ports to allow quick transition to the forwarding state. BPDU Guard is enabled to protect the network from spanning-tree topology disruptions.

```
ASW-B1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-B1(config)#spanning-tree mode rapid-pvst
ASW-B1(config)#interface f0/1
ASW-B1(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
ASW-B1(config-if)#spanning-tree bpduguard enable
ASW-B1(config-if)#exit
ASW-B1(config)#do write
Building configuration...
[OK]
```

- **ASW-B2**
  Rapid PVST+ is enabled and PortFast and BPDU Guard are configured on host-facing interfaces to improve connectivity speed and maintain network stability.

```
ASW-B2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-B2(config)#spanning-tree mode rapid-pvst
ASW-B2(config)#interface f0/1
ASW-B2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
ASW-B2(config-if)#spanning-tree bpduguard enable
ASW-B2(config-if)#exit
ASW-B2(config)#do write
Building configuration...
[OK]
```

- **ASW-B3**
  Rapid PVST+ is enabled and PortFast with BPDU Guard is configured on the access ports. This configuration prevents loops and ensures secure connectivity for end devices.

```
ASW-B3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-B3(config)#spanning-tree mode rapid-pvst
ASW-B3(config)#interface f0/1
ASW-B3(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
ASW-B3(config-if)#spanning-tree bpduguard enable
ASW-B3(config-if)#exit
ASW-B3(config)#do write
Building configuration...
[OK]
```